Functions and Function Notation

Function notation

1 Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0)$ b $f(2)$ c $f(-1)$ d $f\left(\frac{1}{2}\right)$

2 Consider the function $g(x) = \sqrt{x + 5}$. Evaluate:

a $g(4)$ b $g(-1)$ c $g(11)$ d $g(-5)$

3 Consider the function $h(x) = \frac{x - 3}{x + 2}$. Evaluate:

a $h(0)$ b $h(3)$ c $h(-1)$ d $h(5)$

4 If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a.

Identifying functions

5 Determine whether each equation defines y as a function of x:

a $y = 2x + 1$ b $x = y^2$

c $y = |x|$ d $x^2 + y^2 = 9$

6 Explain how the vertical line test is used to decide whether a graph represents a function.

Combining functions

7 Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

a $(f + g)(x)$ b $(f - g)(x)$

c $(f \cdot g)(2)$ d $(f \circ g)(1)$

8 A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.

a Interpret the value $C(0)$ in context.

b Find $C(12)$.

c Solve $C(n) = 925$ and interpret your answer.

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9 Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.

Piecewise-defined functions

10 A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:

a $f(-3)$ **b** $f(0)$ **c** $f(1)$ **d** $f(4)$

11 A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.

a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB.

b Find $C(3)$ and $C(8)$.

c Is C continuous at $x = 5$? Explain.

12 Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.

Even, odd, and one-to-one functions

13 Determine whether each function is even, odd, or neither:

a $f(x) = x^4 - 3x^2$

b $f(x) = x^3 - x$

c $f(x) = x^2 + 2x$

14 Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.

a $f(x) = 2x + 5$ **b** $f(x) = x^2$

Reading function graphs

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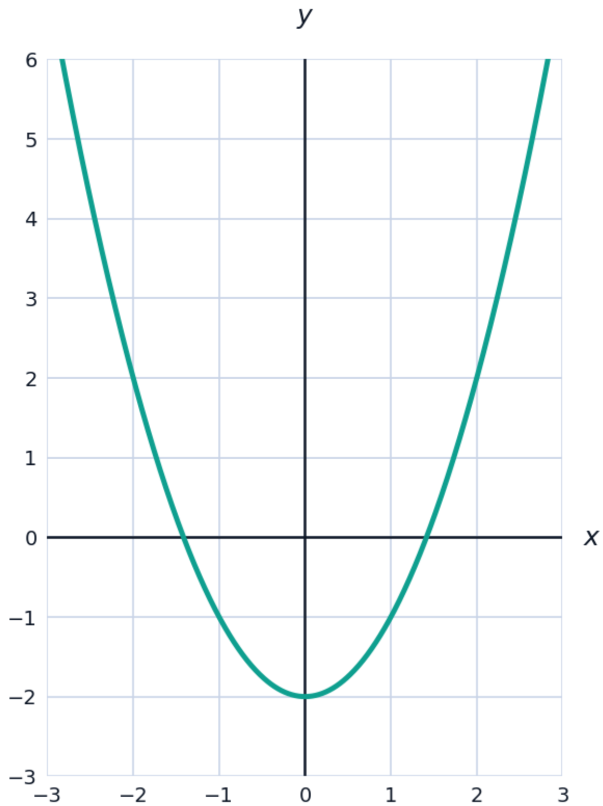
15 The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.



16 Using the graph of $f(x) = x^2 - 2$ above, estimate all x-values where $f(x) = 2$.

Average rate of change from function notation

17 For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.

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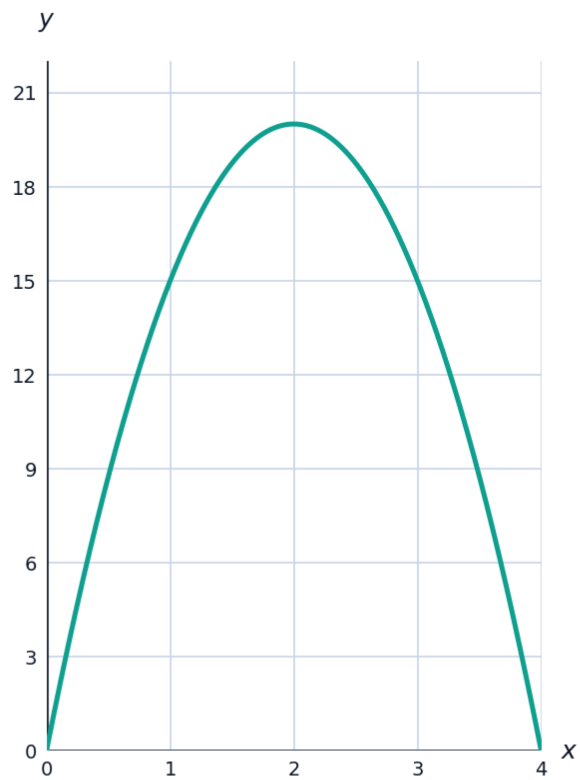
18 A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.



Difference quotients

19 For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$.

20 For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, and explain why the result makes sense for a linear function.

Inverse functions (introduction)

21 Find the inverse of $f(x) = 3x - 4$, and verify by checking $f(f^{-1}(x)) = x$.

22 Explain why $f(x) = x^2$ (with domain all reals) does not have an inverse function, but $f(x) = x^2$ restricted to $x \geq 0$ does.

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